

Math 344, F18
Quiz 8

Name: *Answer*

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Are these pairs of vectors **orthogonal** to each other or only linearly independent?

(a) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} c_1 + c_2 + c_3 \\ c_2 + c_3 \\ c_3 \end{bmatrix} = 0$$

$$\Rightarrow c_1 = c_2 = c_3 = 0 \Rightarrow \text{linearly independent.}$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 1, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 2 \Rightarrow \text{not orthogonal for any of two}$$

(b) $\begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}, \begin{bmatrix} 0.4 \\ 0.3 \end{bmatrix}$.

$$\begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix} \cdot \begin{bmatrix} 0.4 \\ 0.3 \end{bmatrix} \Rightarrow \text{orthogonal}$$

$$= -0.24 + 0.24 = 0$$

(c) $\begin{bmatrix} \frac{1}{2} \cos \theta \\ -\frac{1}{2} \sin \theta \\ \frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} \sin \theta \\ \cos \theta \\ 0 \end{bmatrix}$.

$$\begin{bmatrix} \frac{1}{2} \cos \theta \\ -\frac{1}{2} \sin \theta \\ \frac{\sqrt{3}}{2} \end{bmatrix} \cdot \begin{bmatrix} \sin \theta \\ \cos \theta \\ 0 \end{bmatrix} = \frac{1}{2} \cos \theta \sin \theta - \frac{1}{2} \sin \theta \cos \theta = 0$$

$$\Rightarrow \text{orthogonal}$$

Problem 2. (5 points) Given two vectors $\mathbf{a}_1 = (1, 1, 0)$ and $\mathbf{a}_2 = (1, 1, 1)$.

(a) Find the projection of $\mathbf{b} = (2, 4, 6)$ onto the subspace of $\mathbb{R}^3$ spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$.

$$A = [\mathbf{a}_1, \mathbf{a}_2] = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

$$\text{Solve } A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}, \quad A^T \mathbf{b} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \end{bmatrix} \Rightarrow \hat{\mathbf{x}} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$$

$$\Rightarrow \mathbf{p} = A \hat{\mathbf{x}} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix}$$

(b) Find the projection matrix $\mathbf{P}$ for the subspace of $\mathbb{R}^3$ spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$.

The projection matrix

$$\begin{aligned} \mathbf{P} &= A(A^T A)^{-1} A^T \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \left(\frac{1}{2}\right) \begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$